Assessing Cognitive Workload During Dual-Task Performance: An fNIRS Study of N-back Tasks Under Walking and Standing Conditions

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Synopsis

Brain-Computer Interfaces (BCI) have traditionally focused on detecting users' unconscious brain states in stationary positions. Previous research has demonstrated that walking while performing complex cognitive tasks increases mental workload in the prefrontal cortex (Hoang et al., 2020). This study investigates differences in mental workload between standing and walking conditions using multiple sensor modalities designed to assess cognitive load. Our results indicate that standing conditions led to increased activity in the frontal lobe but did not affect subjective cognitive workload assessments.

Background

Previous work has shown that mental workload in the prefrontal cortex can be classified from fNIRS data using a variety of machine learning models (Benerradi et al., 2023). Functional Near-Infrared Spectroscopy (fNIRS) data has shown activation in the prefrontal cortex (PFC) during cognitive tasks including problem solving, planning, reasoning, and working memory (Bunce et al., 2011).

Methods

Nineteen participants confirmed that they met the eligibility criteria for this study. All participants did not have a history of brain injury or psychological disorder and were not on medication affecting brain activity. Prior to the study, all participants signed consent forms approved by the Social, Behavioral & Educational Research Institutional Review Board (SBER IRB) at Tufts University. Participants were given monetary compensation of $20/hr for their time. Data from 4 of the 19 participants had to be removed from analysis due to device malfunction.

Functional near-infrared spectroscopy (fNIRS) data was recorded using an Artinis Brite device at 14 different optode locations covering the prefrontal cortex and motor cortices in the left and right hemispheres. fNIRS data was recorded using the Artnis Oxysoft software.

Electroencephalography (EEG) data was recorded using the Neuroelectrics Enobio 32 and pupil dilation data was recorded using the SMI ETG2 Eye Tracking Glasses. Due to technical failures, the data from both devices were unusable, and further analysis was done using fNIRS and survey data.

The NASA Task Load Index (NASA-TLX) was used to measure subjective workload at the end of each block. The participant was asked to assess their level of mental workload after completing a single block of an n-back task at a particular difficulty (0, 1, 2). The NASA-TLX is a series of questions that targets six aspects of mental workload on a 21-point Likert scale.

The experiment followed a 3x2 within-subjects design with three workload levels and two movement conditions. Participants stood on a treadmill facing a computer screen at a standing desk as shown in Figure 1. Each block began with a fixation cross during a 30 second rest period, followed by presentation of the n-back level and simultaneous movement condition. Participants completed 40 rounds of the n-back task, with each task block lasting 90 seconds

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(Owen et al., 2005; Jaeggi et al., 2010). After six practice blocks covering all task combinations, participants completed 18 randomly assigned experimental blocks.

The modified Beer-Lambert law and bandpass filtering were applied through Artinis Oxysoft. Baseline correction was also performed on the fNIRS data using the rest periods. Channels were rejected using a paired t-test using the provided function in Homer3 (Huppert et al., 2009). We kept a single channel with only the most significant changes for each ROI to reduce the number of input features for the classification model (Hong et al., 2018; Li et al., 2017).



Figure 1. Participant is shown standing on a treadmill in front of the standing desk setup.

Normality was assessed using Shapiro-Wilk tests at each n-back level for individual channels and regions of interest. Repeated measures ANOVA were performed on HbO during standing to test n-back effects as well as overall movement effects. Pairwise comparisons between standing and walking conditions at each n-back level used paired t-tests or Wilcoxon signed-rank tests depending on the normality of differences. All p-values were corrected with false discovery rate (FDR) correction for multiple comparisons (Benjamini & Hochberg, 1995).

Results

fNIRS analysis showed statistically significant results for the 0-back versus 2-back comparison without walking conditions on a single channel in the left frontal lobe ($F(1, 14) = 13.86$, $p = .032$, $\eta^2_g = 0.131$). A statistically significant change in oxygenated hemoglobin in the right frontal lobe was found between standing and walking conditions during the 0-back ($t(12) = -4.04$, $p = 0.034$, $d = -1.12$) and 2-back tasks ($t(12) = -3.74$, $p = 0.034$, $d = -1.04$). ANOVA results for overall movement comparisons returned a statistically significant result for the frontal right HbO in which higher HbO was found for standing conditions ($F(1,12) = 17.45$, $p = 0.0103$, $\eta^2 = 0.187$). Classification was done using a Gaussian distribution with a decision threshold at 50% which achieved an average F1 score of 66.84% and accuracy of 50.25% on a test set, which is on par with the current standard (Benerradi et al., 2023).

NASA-TLX subjective workload assessments demonstrated statistically significant increases in mental demand for both 1-back and 2-back tasks compared to baseline, though movement conditions did not affect mental demand ratings. Physical demand ratings significantly increased during walking compared to standing but were unaffected by n-back level. Temporal demand showed significant changes based on both n-back level and movement condition independently, with no interaction effect between the two factors. Performance and effort ratings both showed statistically significant changes related to n-back level but were unaffected by walking versus standing conditions.

Discussion

This study demonstrated that cognitive workload and physical activity produce distinct physiological and subjective responses. The significant fNIRS findings in the frontal region during varying N-back levels suggest increased prefrontal activation with higher cognitive demands, consistent with working memory literature (Owen et al., 2005). The independence of task difficulty and physical demand suggest cognitive and physical workload operate through separate mechanisms, although future work is necessary to confirm this. We find that different physical conditions while completing complex cognitive tasks leads to differentiations in oxygenations which should be considered for mobile BCIs.

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